

Phase 6 β Status & Open Questions

where the science is right now (May 2026)

Onboarding doc 04 of 06

May 11, 2026

What this document is for

Maps the recent project history onto the actual code so that the strange-sounding labels in commit messages (“v9”, “v10a”, “Step 9.5”, “Phase D”) connect to what the model is doing and why. Closes with a list of contributions a new collaborator can realistically pick up.

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1 The arc in one paragraph

Phase 6 is the project epoch in which we moved from a *transport-only* model that could not match the experimental local-pH excursions to a *chemistry-augmented* model that accounts for the cation's role in modulating local proton supply. The first sub-phase (Phase 6 α) added water self-ionization ($\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$) and showed it cannot explain the deck data. The second sub-phase (Phase 6 β , ongoing) adds the actual mechanism the experimental group believes in: field-dependent hydrolysis

of hydrated alkali cations at the outer Helmholtz plane (OHP), per Singh 2016. Phase 6 β has gone through several internal revisions (v9, v10a, v10b) and is currently in its first *fitting* stage (Step 10 / Phase D), trying to recover the cation-specific Singh parameters from peroxide-onset data in the Mangan deck.

2 Phase 6 α : water self-ionization (landed)

Question. Does adding $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$ as a homogeneous bulk source/sink in the proton equation lower the predicted surface pH from ~ 14 (Phase 5 baseline) into the deck-observed window of ~ 5 – 7 ?

Answer. Partially, and not enough.

- Surface pH dropped from 14.14 (transport-saturated) to 10.58 with K_w active — a 3.5-unit improvement.
- The 10.58 plateau is independent of the boundary-layer thickness L_{eff} : making the diffusion layer thinner does not lower it further. Surface pH is set by the *kinetic-transport balance*, not by transport alone.
- The textbook bulk water-dissociation rate is $\sim 1.4 \text{ mol/m}^3/\text{s}$; the deficit at $L = 16 \mu\text{m}$ cathodic demands $\sim 30 \text{ mol/m}^3/\text{s}$ — $20\times$ faster than water can actually supply. So the Phase 6 α prediction of 10.58 is itself optimistic; the true gap to the deck pH window is even larger.

Take-away. Water self-ionization is universal aqueous physics and we keep it as an opt-in residual (default-off; flag `enable_water_ionization`). But it is *not* the mechanism that explains cation effects. Documented in `docs/phase6/PHASE_6A_INVESTIGATION_SUMMARY.md`.

3 Phase 6 β : the cation-hydrolysis program

3.1 The hypothesis (recap)

The reaction (cation hydrolysis, doc 01):



with field-dependent pK_a (Singh formula, doc 01):

$$\text{pK}_a(\sigma) = \text{pK}_{a,\text{bulk}} + 2Az\sigma r_{H-El} \left(1 - \frac{r_{M-O}^2}{r_{H-El}^2} \right).$$

This makes the cation a local proton donor whose strength depends on both its identity (bulk pK_a) and the local Stern-layer surface charge σ .

3.2 v9: the architectural skeleton (May 2026, landed)

v9 wired this into the solver. Specifically:

- Promoted K^+ from an analytic Bikerman counter-ion to a *dynamic* NP species (constants `FOUR_SPECIES_LOGC_DYNAMIC_K2SO4` in `scripts/_bv_common.py`); SO_4^{2-} stays analytic. Reason: the Boltzmann profile gives zero NP flux by construction, so the kinetic source from $c_M(0) \cdot k_{\text{hyd}}$ has nothing to drain.
- Added a global Real-element scalar Γ_{MOH} (mol/m²) on the electrode, evolving by the coverage-rate ODE (doc 02, $\dot{\Gamma}$ formula).
- Implemented Singh’s pK_a formula in `Forward/bv_solver/cation_hydrolysis.py` (function `singh_pka_field_dependent`).
- Outer Picard for Γ_{MOH} around the inner Newton solve.

What v9 found. The architecture works, but the v9 formula has *no Langmuir cap* on Γ_{MOH} . Converged solutions at $k_{\text{hyd}} \geq 10^{-3}$ pile up ~ 6 monolayers of MOH^0 at the OHP — physically absurd since you cannot have more than one monolayer of adsorbed species without a second layer beneath it. v9 is therefore a *numerical* achievement, not a physically defensible model. Documented in `docs/phase6/PHASE_6B_V9_PHASES_A_B_RESULTS_2026-05-10.md`.

3.3 v10a: the Langmuir cap (May 2026, landed)

v10a multiplied the hydrolysis-forward rate by $(1 - \Gamma/\Gamma_{\text{max}})$ so Γ saturates at one monolayer worth of MOH coverage:

$$\dot{\Gamma} = k_{\text{hyd}}(1 - \Gamma/\Gamma_{\text{max}}) c_M(0) - k_{\text{des}}\Gamma.$$

This restored physical sensibility and unlocked the next round of calibration questions: *what is Γ_{max} ? what is k_{hyd} ? what is k_{des} ?*

3.4 v10a’ and Phase A.2: pick a calibration voltage

The current-voltage curve has many points; if you fit cation parameters using all 24 points at once, you cannot tell which point the residual is being driven by. v10a’ proposed a **kinetic-anchor voltage** V_{kin} where (i) the current is well in the kinetic-controlled regime (not Levich-limited), (ii) the Stern surface charge is large enough that cation hydrolysis matters, and (iii) the multi-ion solver actually converges from cold-start. The chosen value: $V_{\text{kin}} = -0.10$ V vs. RHE.

Phase A.2 then re-derived a clean baseline at V_{kin} with v10a’s Langmuir cap, providing a high-confidence anchor for later sweeps. Driver: `scripts/studies/phase6b_v10a_phase_A2_v_kin.py`.

3.5 v10b: literature anchoring of constants (May 2026, landed)

v10b set the three free Phase 6 β constants to their literature-anchored values:

Constant	Value	Source
Γ_{max} (nondim)	0.047	V10A chain (tightened); literature compatibility audit (4 cross-tests).
k_{des} (nondim)	1.0	Engineering choice; Eyring prior $[10^{-2}, 10^2] \Leftrightarrow \Delta G_{\text{des}} \in [0.69, 0.94]$ eV.
C_S (F/m ²)	0.20	Bohra–Koper–Choi consensus on CMK-3 carbon.

The constants live in the top-level Firedrake-free `calibration/v10b.py`, importable without the heavy stack. **Deprecation alias:** `SMOKE = V10A_SMOKE` (NEVER `SMOKE = V10B`). Documented in `docs/phase6/v10b_calibration_summary.md`.

3.6 Steps 6–9: ablation, audit, sweeps

After v10b landed, the project enumerated a series of *step-numbered* verification and sweeping tasks that lock in plumbing-level correctness before a fitting attempt:

Step 6: ablation flags. Five default-off knobs in `Forward/bv_solver/config.py` that let you turn individual physics terms on or off (`apply_h_source`, `apply_k_sink`, etc.). All five ablations passed at A.2 baseline.

Step 7: C_S literature lock. `docs/phase6/CMK3_capacitance_literature.md` is the citation chain that pins $C_S = 0.20 \text{ F/m}^2$.

Step 8: missing-data ledger. Open data items the project still needs from the experimental group; tracked in `docs/phase6/missing_data.md`.

Step 9: $k_{\text{hyd}} \times \lambda$ ramp at v10b parameters. A $14 \times 10 = 140$ -rung sweep over hydrolysis rate and the activation parameter $\lambda \in [0, 1]$ at V_{kin} . **Result (2026-05-11):** 14/14 converge with mass balance $\leq 5 \times 10^{-13}$; baseline reproduction byte-equivalent to v10b A.2. Documented in `docs/phase6/phase6b_step9_B2_summary.md`.

Step 9.5: warm_start_floor in AdaptiveLadder. Required to converge the three highest- k_{hyd} extension points in step 9. Opt-in floor; default-off behavior is byte-equivalent. Documented in the same summary.

3.7 Step 10 / Phase D: derivative-free fit of Δ_β (in flight)

This is where the project is right now (May 2026).

Question. Given the v10b-anchored model with all other constants pinned, what value of the free coupling parameter Δ_β (the additive shift to Singh’s $\beta_M = -Azr_{H-El}(1 - r_{M-O}^2/r_{H-El}^2)$ per cation) best reproduces the peroxide-onset voltage in the Mangan deck for K^+ at pH 4?

Status. Plan was iterated through 7 critique rounds (`docs/handoffs/CHATGPT_HANDOFF_37_phase6b-step10-ph` APPROVED at round 7. Two structural corrections in early rounds:

1. the original Δ_β bracket was off by 7 decades (units: β is in pm^2 , not $1/\text{pm}^2$);
2. the original continuation topology was invented; switched to the validated A.2 topology (anchor at $\lambda = 0$, V warm-walk at $\lambda = 0$, per-V λ -ramp $0 \rightarrow 1$). Wall budget grew from 5 hours to 15 hours — overnight execution scope.

Phase D code is in active development:

- `scripts/studies/phase6b_step10_phase_D_orchestrate.py`
- `scripts/studies/phase6b_step10_phase_D_fit_eval.py`
- `scripts/studies/_phase_D_validate_optimized_path.py`
- Tests: `tests/test_phase6b_step10_phase_D_{orchestrate,fit_eval,plumbing}.py`.

The optimizer is `scipy.optimize.minimize_scalar` with `method="bounded"` (Brent’s method), since adjoints through the Picard outer loop are not yet hooked up. The bounded interval in target- ΔpKa -effect space gives a one-sided search respecting the negative- Δ_β sign-guard requirement.

4 What is locked vs. what is open

4.1 Locked (don't change without conversation)

- Production stack: 3 dynamic species + analytic Bikerman counter-ions, `logc_muh`, log-rate BV with parallel 2e/4e at Ruggiero E° .
- $C_S = 0.20$ F/m² (Step 7 lock; CMK-3 literature consensus).
- $\Gamma_{\max} = 0.047$ (nondim) and $k_{\text{des}} = 1.0$ (nondim) (v10b lock).
- Hard rules #1–#6 from CLAUDE.md (anchor-and-grid, `exponent_clip = 100`, IC/residual saturation agreement, parallel 2e/4e topology, Singh 2016 mechanism, C_S value).
- Acceptance bundle (the gate set Phase D will be judged against): primary metric H₂O₂% within ± 10 pp of deck K⁺ pH 4; secondaries on disk current and surface pH envelopes. Documented in `docs/phase6/PHASE_0_ACCEPTANCE_BUNDLE_LOCK_2026-05-10.md`.

4.2 Open

- Phase D fit completion across all four cations (the current scope is K⁺-only first; Li⁺, Na⁺, Cs⁺ are queued).
- Adjoint annotation through the cation-hydrolysis Picard loop (currently a derivative-free fit; to be replaced when the inverse pipeline restarts).
- Phase 6 α .1: a finite-rate water-dissociation refinement, since textbook K_w relaxation is too slow to feed the surface at the rate the equilibrium model assumes. Queued, not started.
- Per-cation Tafel slope extraction from Brianna's 2019 LSV data (`scripts/derive/extract_k_plus_tafel_slope` K⁺ done, others pending).
- Comparison plotting between simulated and deck observables (currently ad-hoc; no canonical figure-generation script).
- Test coverage for the new Phase D helpers (`tests/test_phase6b_step10_phase_D_*.py` are scaffolds; flesh out parameterizations).

5 Where a new contributor can plug in

In rough increasing order of scope:

1. **Read the chronology and ask questions.** You are coming in mid-stream; spend the first week on docs 00–04 and the acceptance-bundle lock. Bring a list of things you don't understand to our next 1:1.
2. **Run the Phase A.2 baseline yourself.** Activate the Firedrake venv (doc 05), run `phase6b_v10a_phase_A2_v10a`, look at the JSON and the plot. Verify it matches the existing `StudyResults/` subdirectory.
3. **Add a small diagnostic.** For example: a plot of Γ_{MOH} vs. V_{RHE} overlaid on the BV-rate contributions, or a histogram of $\text{pK}_a(\sigma)$ over the deck V-band. The pattern is in doc 03 §6.

4. **Reproduce a Singh table number.** Pick a row of the Singh Table S1 (Li, Na, K, Rb, Cs); plug r_M and z into the bulk pK_a formula (Singh formula with $\sigma = 0$); verify our `calibration/singh2016.py` returns the same number to four decimal places. This is exactly the kind of numerical-verification test the project values most.
5. **Improve a regression test.** The Phase D scaffolding tests are minimal; pick one and add three or four property-based parameterizations (e.g. “ $\Delta_\beta = 0$ should reproduce the v10b A.2 baseline byte-for-byte”; “ $\Delta_\beta \rightarrow \infty$ should saturate Γ_{MOH} at Γ_{max} ”; etc.).
6. **Independently audit a Phase D handoff round.** The seven-round critique on the Phase D plan is checked into `docs/handoffs/CHATGPT_HANDOFF_37_*`. Read R1 and R2; can you spot the dimensional-bracket error *before* reading R1’s accepted issues? This trains you to read the acceptance ledgers.

After 4–6 weeks of these we will pick something more ambitious from the “open” list above.